

# Department of Computer Science Engineering

## UE23CS352A: Machine Learning Hackathon

# Hackman

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## Analysis Report

### Key Observations:

The project presented several notable challenges. One of the most significant was selecting letters in a way that minimized incorrect guesses within the fixed limit imposed by the Hangman rules. The Hidden Markov Model (HMM) implementation proved essential for narrowing down the search space of possible words, particularly in the early stages of the game. However, achieving precise pattern matching and balancing global letter frequencies required careful tuning.

The reinforcement learning (RL) component—based on a Deep Q-Network (DQN)—faced its own difficulties, particularly in dealing with sparse rewards and a relatively large state space. The RL agent had to make decisions based not only on the revealed pattern and guesses but also on additional probabilistic cues provided by the HMM. A key insight was that a hybrid approach, where the HMM guided early guesses and the RL agent refined decisions later, resulted in improved overall performance.

### Strategies:

#### 1. Design of the Hidden Markov Model (HMM)

- The word corpus was organized by word length so that only valid-length candidates were considered during prediction.

- A global frequency distribution of letters across the corpus was used as a fallback in cases where the model could not confidently filter words based on the revealed pattern.
- A multi-step filtering approach was applied to exclude words that did not match the current pattern or contained letters that had already been ruled out by prior guesses.

## **2. Reinforcement Learning: State and Reward Design**

The agent's state vector consisted of 81 numerical features, including:

- The normalized length of the word and remaining lives.
- Binary indicators representing guessed letters.
- Probabilities obtained from the HMM for each letter.
- Frequency of each letter in the masked word.
- The number of unrevealed characters.

## **3. The reward structure was crafted to strongly discourage repeated or incorrect guesses:**

- Correct guess: +15
- Incorrect guess: -25 (later adjusted to -100 for RL training)
- Repeated guess: -10 (later adjusted to -50)
- Game won: +100
- Game lost: -150

This reward shaping encouraged the model to maximize efficiency in guessing.

## **Exploration–Exploitation Trade-off:**

The RL agent employed an  $\epsilon$ -greedy strategy to balance exploration and exploitation. The  $\epsilon$ -value began at 0.2 and decayed gradually to 0.01, enabling sufficient exploration during the early stages of training. To incorporate domain-specific knowledge, a weighted combination of HMM predictions and Q-values was used when selecting actions, with 70% of the influence initially given to the HMM. This method allowed the agent to explore less frequently and more intelligently, especially in uncertain contexts.

## Future Improvements:

- **Integration of subword information (e.g., bigram or trigram dependencies)** to improve the HMM's predictive capability for incomplete patterns.
- **Use of more advanced RL architectures**, such as Double DQN or Dueling DQN, to reduce overestimation in Q-values and stabilize training.
- **Implementation of prioritized experience replay** to give greater emphasis to more informative transitions during learning.
- **Incorporation of sequence models (e.g., RNNs)** to better retain the context of past guesses over time.
- **Curriculum-based training**, beginning with shorter or more common words and progressively advancing to more challenging vocabulary.